

SENATE No. 2243

The Commonwealth of Massachusetts

PRESENTED BY:

Cynthia Stone Creem

To the Honorable Senate and House of Representatives of the Commonwealth of Massachusetts in General Court assembled:

The undersigned legislators and/or citizens respectfully petition for the adoption of the accompanying bill:

An Act to improve outdoor lighting, conserve energy, and increase dark-sky visibility.

PETITION OF:

NAME:	DISTRICT/ADDRESS:	
<i>Cynthia Stone Creem</i>	<i>Norfolk and Middlesex</i>	
<i>Joanne M. Comerford</i>	<i>Hampshire, Franklin and Worcester</i>	<i>1/31/2025</i>
<i>James B. Eldridge</i>	<i>Middlesex and Worcester</i>	<i>2/10/2025</i>
<i>Bruce E. Tarr</i>	<i>First Essex and Middlesex</i>	<i>2/26/2025</i>
<i>Margaret R. Scarsdale</i>	<i>1st Middlesex</i>	<i>3/17/2025</i>
<i>Edward J. Kennedy</i>	<i>First Middlesex</i>	<i>4/17/2025</i>

SENATE No. 2243

By Ms. Creem, a petition (accompanied by bill, Senate, No. 2243) of Cynthia Stone Creem, Joanne M. Comerford, James B. Eldridge and Bruce E. Tarr for legislation to promote energy efficient lighting, conserve energy, regulate outdoor night lighting, and reduce light pollution. Telecommunications, Utilities and Energy.

[SIMILAR MATTER FILED IN PREVIOUS SESSION
SEE SENATE, NO. 2102 OF 2023-2024.]

The Commonwealth of Massachusetts

In the One Hundred and Ninety-Fourth General Court
(2025-2026)

An Act to improve outdoor lighting, conserve energy, and increase dark-sky visibility.

Be it enacted by the Senate and House of Representatives in General Court assembled, and by the authority of the same, as follows:

1 SECTION 1. Chapter 85 of the General Laws is hereby amended by adding the following
2 section:

3 Section 38. (a) As used in this section, the following words shall have the following
4 meanings unless the context clearly requires otherwise:

5 “Correlated color temperature” or “CCT”, the apparent hue of the light emitted by a
6 fixture, expressed in kelvin

7 (K).

“Façade lighting”, illumination of exterior surfaces of buildings for the enhancement of their nighttime appearance, achieved by shining light onto building surfaces, or by internal or external illumination of translucent building surfaces, or with fixtures solely for decorative function.

“Fixture”, a complete lighting unit, including a light source together with the parts designed to distribute the light, position and protect the light source, and connect the light source to the power supply.

“Fully shielded fixture”, a fixture that in its mounted position has an uplight value of U0 as defined by the standards of the Illuminating Engineering Society or that allows no direct light from the fixture above a horizontal plane through the fixture's lowest light-emitting part, in its mounted position.

“Glare”, light emitted by a fixture that causes visual discomfort or reduced visibility.

“Illuminance”, the luminous power incident per unit area of a surface.

“Light trespass”, light that falls beyond the property it is intended to illuminate.

“Lumen”, a standard unit of measurement of the quantity of light emitted from a source of light.

“Municipal funds”, bond revenues or money appropriated or allocated by the governing body of a town or city within the commonwealth.

“Ornamental lighting”, a lighting fixture that has a historical or decorative appearance and that serves a decorative function in addition to serving to light a roadway, parking lot, walkway, plaza, landscaping, or other area.

“Parking-lot lighting”, a permanent outdoor fixture specifically intended to illuminate an uncovered

vehicle parking area.

“Permanent outdoor fixture”, a fixture for use in an exterior environment installed with mounting not intended

for relocation.

“Roadway lighting”, a permanent outdoor fixture specifically intended to illuminate a public roadway.

“Sky glow”, scattered light in the atmosphere that is caused by light directed upward or sideways from fixtures, reducing an individual’s ability to view the natural night sky.

“State funds”, bond revenues or money appropriated or allocated by the general court.

“Uplight,” direct light emitted above a horizontal plane through the fixture’s lowest light-emitting part in its

mounted position.

(b) Not later than January 1, 2026, the Department of Energy Resources, in consultation with the Massachusetts Department of Transportation and the Executive Office of Public Safety and Security, shall promulgate regulations with the intent of reducing energy waste and

minimizing light pollution to the greatest extent possible and therefore to ensure that state or municipal funds are not used to install, to cause to be installed, or to operate a new permanent outdoor fixture, unless the following conditions are met:

(i) Fixtures used for roadway lighting or parking-lot lighting, whether mounted to poles, buildings or other structures, shall be fully shielded unless they are ornamental lighting fixtures, or are fixtures used to light tunnels or roadway underpasses;

(ii) Ornamental lighting fixtures shall comply with a limit on lumens of uplight established in the regulations;

(iii) Fixtures used for roadway lighting shall not be more numerous than is necessary for adequate vehicular and pedestrian safety, as determined by the regulations and given due consideration to the current lighting needs criteria published by the Federal Highway Administration and the Illuminating Engineering Society;

(iv) Building-mounted fixtures shall be fully shielded unless they are façade lighting fixtures;

(v) Façade lighting fixtures shall be selected and installed to direct the light onto the intended target, and shall be shielded, so that glare, sky glow, and light trespass are minimized;

(vi) Fixtures used to light historic structures, flags, monuments, statuary and works of art shall be selected and installed to direct the light onto the intended target, and shall be shielded, so that glare, sky glow, and light trespass are minimized;

(vii) Fixtures used to light athletic playing areas shall be selected and installed so as to minimize glare, light trespass and sky glow outside the athletic playing area;

(viii) Fixtures installed for any purpose shall have a correlated color temperature that is not greater than a limit established in the regulations; provided, that the limit shall not exceed 3000 K; and provided further, that said regulations shall include exemptions from the limit where a public safety need is demonstrated, where the fixtures are used exclusively for the decorative illumination through color of certain building façade or landscape features, or where the fixtures are used to illuminate athletic playing areas;

(ix) Lighting installed for any purpose shall provide maintained illuminance at levels that are no greater than those required for the intended purpose, as established in the regulations and given due consideration to lighting industry standards and practices, unless a demonstrated and verified need exists for higher levels to ensure safety or security; provided, that if a municipal or county ordinance or regulation specifies lower illuminance levels, the illuminance level required for the intended purpose by the municipal or county ordinance or regulation shall be used.

(c) This section shall not apply: (i) if it is preempted by federal law; (ii) if the outdoor lighting fixture is used temporarily for emergency, repair, construction or similar activities; (iii) to navigational and other lighting systems necessary for aviation and nautical safety; (iv) if a compelling and bona fide safety or security need exists that cannot be addressed by another reasonable method; (v) to the replacement of a previously installed permanent outdoor fixture that is destroyed, damaged or inoperative, has experienced electrical failure due to failed components, or requires standard maintenance; (vi) to festoon lighting; as defined in the NFPA 70 National Electrical Code; or (vii) to fixtures installed under any specific exemptions as codified in the regulations established under Section 38 (b).

SECTION 2. The Massachusetts Department of Transportation shall review and issue a report on existing roadway lighting and lighting operational costs. The report shall include a review of standards and other criteria for roadway lighting and an analysis of lighting operational costs; a review of roadway lighting's impact on human health, human safety, and environmental impact; actions taken by the department to comply with current standards; procedures and accepted best practices relative to roadway lighting; and recommendations for reducing lighting operational costs through the replacement of existing high-wattage, unshielded fixtures with lower-wattage, fully shielded fixtures and the replacement of unnecessary roadway lighting with the installation of passive safety measures. The department shall issue its report to the department of energy resources and the clerks of senate and the house of representatives not later than January 1, 2026.

SECTION 3. Not later than January 1, 2026, the Department of Public Utilities shall, subject to its ratemaking authority, develop a rate for unmetered roadway or parking-lot lighting fixtures utilizing less than 25 watts of electricity.